

120 Days of Quantitative Finance

Month 4: Stochastic Calculus & Time Series Modelling

Week 15: Advanced Time Series Modelling

Day 102: ARIMA Models and Model Selection

1 Introduction

ARIMA models extend ARMA models to handle non-stationary time series by incorporating differencing.

They are widely used in financial forecasting.

2 ARIMA Model Definition

An $\text{ARIMA}(p, d, q)$ model consists of:

- p : autoregressive order
- d : degree of differencing
- q : moving average order

After differencing d times, the series becomes stationary and can be modeled using $\text{ARMA}(p, q)$.

3 Mathematical Representation

$$\Delta^d X_t = c + \sum_{i=1}^p \phi_i \Delta^d X_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t$$

4 Model Selection Problem

Choosing (p, d, q) is critical.

Overfitting leads to poor generalization, while underfitting misses structure.

5 Akaike Information Criterion (AIC)

$$AIC = -2\log(L) + 2k$$

where:

- L is likelihood
- k is number of parameters

6 Bayesian Information Criterion (BIC)

$$BIC = -2\log(L) + k\log(n)$$

7 Interpretation

- Lower AIC/BIC indicates better model
- AIC favors better fit
- BIC penalizes complexity more strongly

8 Trade-Off

- Fit vs simplicity
- Bias vs variance

9 Financial Relevance

ARIMA models are used to:

- forecast returns

- model macroeconomic time series
- detect patterns in financial data

10 Conclusion

ARIMA provides a flexible framework for modeling time series, while AIC and BIC help select the most appropriate model objectively.